

V MUKESH KUMAR

Software Engineer | Unreal Engine (C++) | Gameplay Systems, GAS & Multiplayer

Bengaluru, Karnataka, India | +91 96320 48141 | v.mukesh00kumar@gmail.com

[linkedin.com/in/v-mukesh-kumar-829aa2173](https://www.linkedin.com/in/v-mukesh-kumar-829aa2173) | vmukeshk.github.io/mukesh-portfolio

SUMMARY

Software Engineer with **4+ years** building games and real-time simulations in **Unreal Engine (UE4/UE5)** and **Unity**, shipped to Android, iOS and Linux. **Frontend Lead on four projects**, owning gameplay architecture, estimation, sprint planning and delivery. Deep hands-on experience with the **Gameplay Ability System**, modular gameplay patterns, **server-authoritative multiplayer** with replication and prediction, and animation, camera and AI systems. Built custom engine plugins with native Android and iOS layers when the engine didn't offer what the project needed. Strong C++/C# fundamentals and a track record of learning complex domains independently and shipping production-ready systems.

SKILLS

Engines & Languages: Unreal Engine 4/5 (C++, Blueprints), Unity (C#, ScriptableObjects, Addressables), Java, Objective-C, SOLID / DRY, event-driven and modular architecture

Gameplay Systems: Gameplay Ability System (GAS), Gameplay Effects / Cues, Modular Gameplay & Game Feature Plugins, Asset Manager, Gameplay Message Subsystem, Slate / UMG, Behavior Trees & EQS, Camera Modes, Linked Anim Layers, GASP, Niagara, PCG

Multiplayer & Networking: Server-authoritative gameplay, replication & prediction, dedicated servers, sessions, lobbies, matchmaking, latency handling

Debugging & Performance: Unreal Insights, ADB / logcat, crash triage, profiling and optimization, build & link failure diagnosis

Plugin & SDK Development: Custom Unreal plugins and runtime/editor modules (.uplugin, .Build.cs, UPL), Game Feature plugins, Unity packages and native plugins, API design for consuming teams, engine-version compatibility, integration docs

Native Mobile Integration: Android (JNI, Java, Gradle/AAR, UPL, scoped storage / API 33+), iOS (Objective-C, Xcode build phases), native-to-engine callback bridging, lifecycle handling

Third-Party SDKs: In-App Purchases, AdMob, PlayFab, social sign-in (Google, Apple, Facebook), analytics, cloud save, REST / WebSocket APIs, Photon Realtime, Online Subsystems (Steam, EOS), CryptoPP / OpenSSL

Tools & Workflow: Git, Unreal Editor tooling, Pixel Streaming, AWS, Agile / Scrum estimation and sprint planning, AI-assisted development (Claude and similar tools) for code, review and documentation

EXPERIENCE

Juego Studios

Jul 2022 – Present • 4 yrs 3 mos

Software Engineer II • Bengaluru, Karnataka • Promoted three times in three years

Software Engineer II (Nov 2025 – Present) • Software Engineer I (Oct 2023 – Oct 2025) • Trainee Developer (Nov 2022 – Oct 2023) •

Intern Developer (Jul 2022 – Oct 2022)

BAM (By Any Means) — 4v4 Tactical Shooter / MOBA Hybrid

Frontend Lead • Unreal Engine 5.8 • Closed Alpha

- Frontend Lead for a competitive 4v4 multiplayer hero shooter, now in **closed alpha with internal playtests**; lead a frontend team of up to **6 engineers** and coordinate across a ~20-person cross-functional team of design, art, backend and QA.
- Designed four hero classes (Rifleman, Grenadier, Breacher, Negotiator) on the **Gameplay Ability System** inside a modular **Game Feature plugin** — abilities, costs, cooldowns, cues — with **server-authoritative logic, replication and latency handling**.
- Built the component-driven experience flow: 4v4 round transitions and client-selection RPCs in a mode component, startup routing through a custom **Asset Manager** loading bundles and action sets per experience, and UI replication over the **Gameplay Message Subsystem**.
- Integrated backend matchmaking and session services (REST APIs and sockets) into the Unreal client and shipped dedicated server builds for **AWS GameLift**-hosted matches.
- Diagnosed and resolved **replication and performance issues** surfaced by playtests, and led level-design optimization for the competitive arena map.
- Built **PCG**-driven modular environment tooling (procedural wall and rampart assembly); integrating **GASP** motion-matching locomotion with class-specific ability and melee animations.

- Co-authored the competitive map design doc — research-facility arena, movement metrics, ability cast times and VFX budgets — and collaborate with designers and backend on balance, progression and telemetry-driven tuning.

Craftlife (Android / iOS)

Frontend Lead · Unreal Engine 5

- Delivered the studio's **first Unreal Engine 5 mobile client end-to-end** with a small team; owned frontend architecture, estimation, milestone planning and sprint execution.
- Designed and shipped **reusable Unreal plugins consumed by the project team**: Google Sign-In (JNI, UPL, Java), native Android share sheet, and a PlayFab API integration layer.
- Diagnosed and fixed **engine-level Android crashes** caused by scoped-storage changes (API 33+) by modifying UPL logic and generated Java; validated with ADB and Unreal Insights.
- Integrated **In-App Purchases**, analytics and cloud save with cross-device progression, and built crafting, mining and harvesting loops, a modular quest system, guilds, marketplace, leaderboards and chat over WebSocket and HTTP, in line with Google Play and App Store guidelines.

Musical DNA — Real-Time Music Visualization (Android / iOS)

Frontend Lead · Unreal Engine 5.7

- Led frontend gameplay, UI and MIDI integration for a mobile music visualization game that turns live MIDI note events into Niagara VFX.
- Built **custom Unreal plugins for real-time MIDI input** on Android (JNI, UPL, Java) and iOS (Objective-C), with platform-specific routing and per-channel state components bridging native callbacks into the engine.
- Implemented real-time music-theory classifiers in C++ (note, scale, chord, inversion and shape detection) against scale and chord databases; a visualizer factory subsystem spawning and laying out **Niagara** visualizers; and a file metadata system for recording and replaying sessions.
- Developed REST API handlers, **social sign-in (Google, Apple, Facebook)** and WebSocket connections for the mobile backend.
- Owned estimation, sprint planning, task assignment and cross-team coordination; reached multiple milestones before the project was paused for client reasons.

Claryo — Warehouse Simulation & Training Tool (Linux)

Frontend Lead · Unreal Engine

- Led frontend development for a **team of four**; owned initial estimation, milestone and sprint planning, and task assignment.
- Developed a **QR / barcode scanning plugin** by integrating third-party libraries with Linux support, packaged as a reusable Unreal plugin.
- Implemented level-creation workflows, save/load, multi-camera viewing, runtime actor transform editing, and forklift and worker NPC AI.
- Primary technical point of contact on client calls; **delivered on time** without slips.

Additional Unreal Projects

VR · Multiplayer · Backend Integration

- **JVerse VR** — multiplayer over Photon Realtime SDK, voice chat, VR UI system, GAS-based interactions; build support for Oculus and Pico. **Gummy Bears** — multiplayer for player, monster AI, projectiles, UI and VFX.
- **Clan System** — HTTP and WebSocket communication via engine modules with **AES encryption (CryptoPP, OpenSSL)**. **Biobrx** — interaction system, runtime gizmo, drag-and-drop UI, runtime skeletal-mesh character generator, Discord bot bridge.
- VR training simulator (Chaos vehicles, task-flow module, UI); Pixel Streaming for a large world map with MetaHumans on an **AWS** Linux server; Swarm AI with flanking, team behaviors and **EQS**.

Unity Projects

C# · Mobile

- **My Little Pony: Story Creator** (PlayDate Digital, shipped on Google Play) — designed the content system: **Addressables**-loaded pages, layouts and assets driven by **ScriptableObject**s with animated pagination, so junior developers added pages by entering data rather than writing code.
- **Kingpin Ludo** — REST and WebSocket integration, force-update flow, **AdMob**, performance optimization. **Fashion Experience** — mission and reward systems, UI animation. **Verse / KidVerse** — Unity AI agents, vehicle AI, grid placement, voice-over integration. **Strange Clan / Monster Defense** — attack and tower VFX, REST APIs.

EDUCATION

B.E. Electronics and Communication Engineering · Sahyadri College of Engineering and Management, Mangalore **2022**

LANGUAGES

English, Kannada, Tamil